ST JOSEPH ENGINEERING COLLEGE
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Mini - Project Report on

"DRY SOLAR PANEL CLEANING SYSTEM"

Submitted in partial fulfillment of the requirements for the award of the degree

BACHELOR OF ENGINEERING
in
ELECTRONICS AND COMMUNICATION ENGINEERING

By

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ST JOSEPH ENGINEERING COLLEGE

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DEPARTMENT OF
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CERTIFICATE

Certified that the mini-project work entitled "**Dry Solar Panel Cleaning Syatems**" is carried out by **Akarshan V H, B Vikas, Deevith Hegde and Manish** of USN **4SO21EC007**, **4SO21EC014**, **4SO21EC019** and **4SO21EC056**, a bonafide students at **St Joseph Engineering College, Mangaluru**, in partial fulfillment for the award of degree **Bachelor of Engineering** in **Electronics And Communication Engineering** from **Visvesvaraya Technological University, Belagavi** during the academic year 2023-2024. It is certified that all the correction/suggestions indicated during the interim evaluation have been incorporated in the report. The mini-project report has been approved as it satisfies the academic requirements in respect of the mini-project work prescribed for the said degree.

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DECLARATION

We, the students of Sixth semester, Department of Electronics And Communication Engineering, St Joseph Engineering College, Mangaluru, hereby declare that the work being presented in the dissertation titled "**Dry Solar Panel Cleaning System**" is an authentic record of the work that has been carried out under the guidance of **Ms. Vijayalaxmi H Manjunatha, Assistant Professor.** This mini-project work is submitted in partial fulfillment of the requirements for the award of the degree - Bachelor of Engineering in Electronics And Communication Engineering during the academic year 2023 – 2024. Further the matter embodied in the report has not been submitted in part or full to any other University, Institution or Professional body for the award of any degree or diploma.

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ABSTRACT

Solar energy is the most abundant source of energy. But the solar technology has not matured to the extent of the conventional sources of energy. It faces lots of challenges such as accumulation of dust on the surface of solar panel which leads to reduction in panel output and overall panel efficiency. In order to overcome this problem, the automated cleaning system is developed. This will help arrays of solar panel to achieve a production output closer to their maximum potential and save companies on cost associated energy generation.

The effects of dust on solar panels and cleaning system is well reported and it is found that the power output of the panel is reduced by 50% if the module is not cleaned for a month in tropical countries like India. Therefore automated cleaning system has been developed using wipers. Similar mechanism is used in automated solar panel cleaning and maintenance system.

The solar panel cleaning system employs geared DC motors operating at 30 RPM to drive a sequence of cleaning components along the panel's length. A rotating brush, powered by a separate 60 RPM motor, sweeps away debris from the panel surface. Following this, a vacuum cleaner removes loose particles, while a microfibre cloth provides a final wipe for a fine finish. This automated process ensures thorough cleaning without damaging the sensitive PV surfaces, enhancing energy efficiency and prolonging the lifespan of solar installations. The automated solar panel cleaning system overcomes the human involvement in cleaning the panel. Hence increasing the efficiency by about 13.23%. The output of 100W solar panel with dust accumulated on it was 13.33 watt which got improved to 15.27 watt after employing the automated cleaning system. Cost of the designed system comes up to rupees 9000.

Thus, the maintenance cost gets reduced as human interference is not required for cleaning purpose thus the system can be deployed in solar power plants and even the homes can be lightened using an automated Solar Panels.

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CHAPTER 1

INTRODUCTION

Solar energy has emerged as a optimum solution for a sustainable energy sources due to its abundant availability and environmental benefits. Solar photovoltaic (PV) panels are at the forefront of this revolution, converting sunlight into electricity with remarkable efficiency. However, their performance can be significantly hindered by the accumulation of dust, dirt, and other particulate matter on their surfaces. This necessitates regular cleaning to maintain optimal efficiency and prolong the panels' lifespan.

1.1 Significance of Solar Panel Cleaning

Efficiency is paramount in solar energy production. Studies have shown that dirty solar panelscan experience up to a 25% reduction in efficiency due to light obstruction by dust and grime (M. Fathima et al., 2018). In regions with arid climates or high pollution levels, this effect can be even more pronounced, leading to substantial losses in energy generation potential over time. Therefore, implementing an effective cleaning system is crucial not only for maximizing energy output but also for ensuring the economic viability of solar power installations.

Moreover, regular maintenance contributes to the reliability and longevity of solar panels. By reducing the buildup of debris, cleaning mitigates the risk of surface corrosion and mechanicaldamage, which can compromise the panels' structural integrity and performance. In commercial and utility-scale solar installations, where profitability hinges on consistent energy production, proactive cleaning strategies are indispensable.

1.2 Overview of the Proposed Cleaning System

The proposed solar panel cleaning system leverages a combination of advanced technologies to streamline the maintenance process and enhance cleaning efficiency. This system integratesthe use of vacuum cleaners, microfibre cloths, and rotating brushes designed to traverse the length of the solar panels. Each component plays a crucial role in ensuring thorough cleaningwithout causing damage to the sensitive PV surfaces.

By combining these elements into a cohesive cleaning system, our project aims to offer a practical solution that addresses the challenges associated with solar panel maintenance. The integration of automated cleaning technologies not only reduces manual labor and operational costs but also enhances cleaning consistency and efficiency, thereby maximizing the energy yield and longevity of solar PV installations.

In conclusion, the development of an efficient solar panel cleaning system is pivotal in optimizing energy production, reducing maintenance costs, and promoting the widespread adoption of solar technology. This project report will delve deeper into the design, implementation, and testing phases of our proposed cleaning system, highlighting its effectiveness and practical applicability in real-world solar installation

CHAPTER 2

LITERATURE SURVEY

2.1 Problem definition:

- Develop a solar panel cleaning system with minimal human intervention, without compensating the quality of the original solar panel.
- Design a basic prototype for the cleaning mechanism that runs across the lengths of the panels
- Facilitate the cleaning of solar panels in water limited regions.

2.2 Objectives:

- The overall efficiency of solar panels decreases due to the accumulation of dust on it's surface.
- Yearly 10-30 billion gallons of water is used worldwide to clean these solar panels which could be an issue in water limited regions.
- Hence, an alternative approach to facilitate the cleaning of solar panels is beneficial.
- The overall efficiency of solar panels decreases due to the accumulation of dust on it's surface.

2.1 A survey on the existing systems

Table 2.1 An overview of the literature survey to identify the gaps

Sl. No.	Title of the Paper, Name of the Journal, Year	Methodology/ Major contributions	Remarks/ Gap identified
1	Electrostatic dust removal using adsorbed moisture assisted charge induction, IEEE Journal of Photovoltaics,2022	An electrode in the formof metal bar passes over the panel surface charging durst particles with an electrical charge.	The vibration of dust particles may cause residual effect of dust particles on the solar panel.

2	Automatic solar panel cleaning using Wiper, International Journal of Multi-Disciplinary research,2023	The document outlines an automatic solar panel cleaning system employing diagonal wipers, powered by a DC motor.	The document lacks details on uncleaned area due to wiper design and doesn't quantify improvement.
3	Solar Powered PV Panel Cleaning Robot , International Conference on Recent Trends on Electronics, Information, Communication & Technology (RTEICT),2020	Research focuses on an autonomous solar panel cleaning robot with multi-stage cleaning: air blowers, liquid application, wiper, and cylindrical brush	Absence of efficiency comparison with manual cleaning methods undermines research validity and comprehensiveness.
4	Automatic Solar Panel Cleaning System Based on Arduino for Dust Removal, International Conference on Artificial Intelligence and Smart Systems, 2021.	It utilizes an exhaust fan as an air blower to dislodge loose dust, followed by a soft cloth wiper to remove remaining debris.	Research lacks testing of various dust types and extreme weather scenarioto assess efficiency and durability.

CHAPTER 3
METHODOLOGY

- Conduct qualitative analysis on available solar panel cleaning techniques and develop a cleaning method without the use of water.
- Confirm technical feasibility without water and integration of vacuum cleaner, brush and a microfiber cloth to clean solar panels.
- Choose components with required specification and build prototype with selected components.
- Evaluate prototype performance on various solar panel surfaces and configurations.
- Finalize design based on test outcomes.

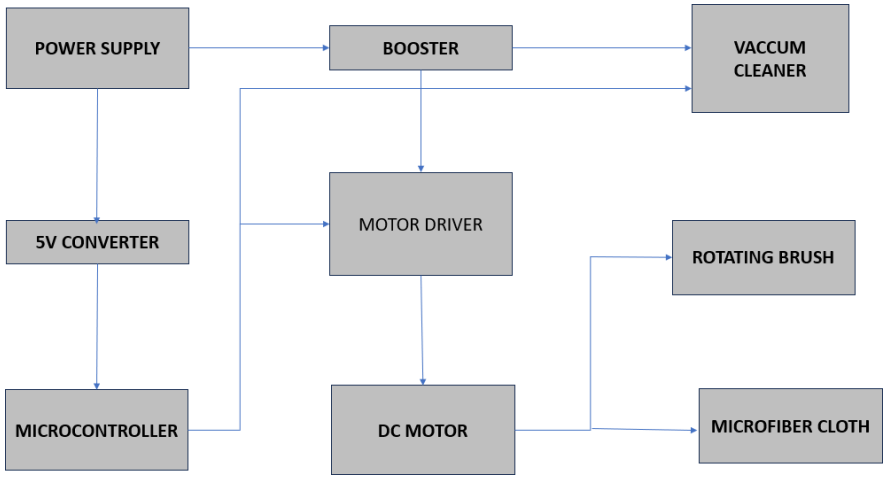


Figure 3.0: Block Diagram

3.1 Research Design

The solar panel cleaning system employs geared DC motors operating at 30 RPM to drive a sequence of cleaning components along the panel's length. A rotating brush, powered by a

separate 60 RPM motor, sweeps away debris from the panel surface. Following this, a vacuum cleaner removes loose particles, while a microfibre cloth provides a final wipe for a fine finish. This automated process ensures thorough cleaning without damaging the sensitive PV surfaces, enhancing energy efficiency and prolonging the lifespan of solar installations. The choice of geared DC motors operating at 30 RPM for driving the cleaning components rotating brush, vacuum cleaner, and microfibre cloth—was carefully considered for several reasons. Firstly, the 30 RPM speed strikes a balance between effective cleaning and gentle treatment of the delicate solar panel surfaces, minimizing the risk of damage. Secondly, the use of geared motors provides precise control over the movement of each component, ensuring thorough coverage and consistent cleaning performance across the entire panel area. This design not only optimizes cleaning efficiency but also enhances the reliability and longevity of the solar panels, supporting sustained energy generation over their operational lifetime.

3.2 Data collection and analysis

3.2.1 Field Visits

The first field visit that we had was at the solar panel installation at St Joseph Engineering College, Manglore. We had an effective communication with the maintenance staff of these solar panels and discussed about their methods and frequencies for cleaning thesepanels. Also received more insights about the maintenance cost which we took into consideration for the financial planning of our project. We also investigated the feasibility of solar panel cleaning systems in this region. This includes assessing the environmental conditions suitable for their operation and identifying potential difficulties encountered during the cleaning process. We will also explore alternative cleaning methods as a comparison. By examining these factors, we can determine the project's overall usefulness and applicability int this specific location.



Figure 3.2.1 SJEC Solar Panel Installation



Figure 2.2NMPT Solar Panel Insstallation

3.3 The Base Circuit:

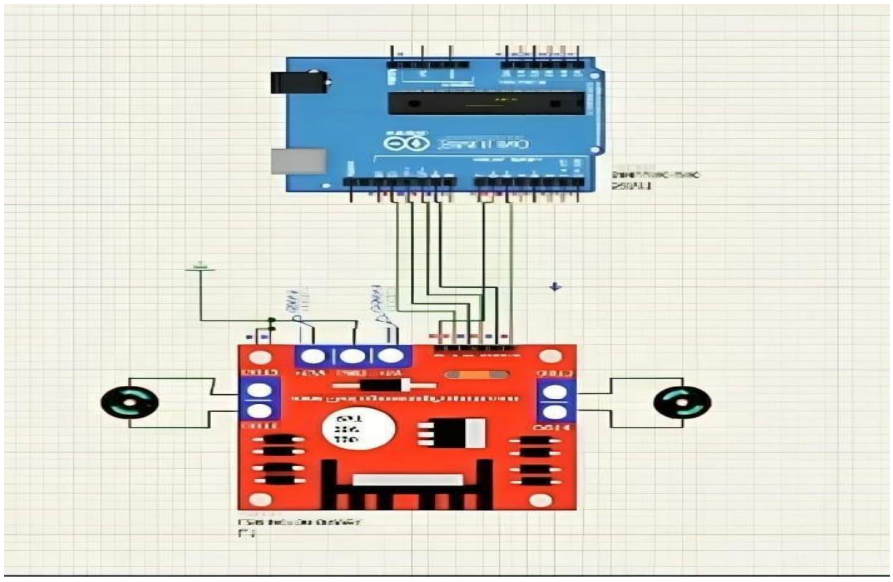


Figure 3.3.1 Circuit diagram

The circuit allows to control a DC motor using an Arduino Uno and an L298D motor driver. The Arduino Uno is responsible for sending control signals, but it can't provide enough power directly to the motor. That's where the L298D comes in. It acts like a bridge, allowing the Arduino's control signals to regulate the higher currents needed by the motor. One power.supply provides voltage to the motor, while another supplies power to the Arduino itself. The Arduino sends direction and speed signals to the L293D, which then controls the motor's direction and speed accordingly.

3.4 Experimental Setup

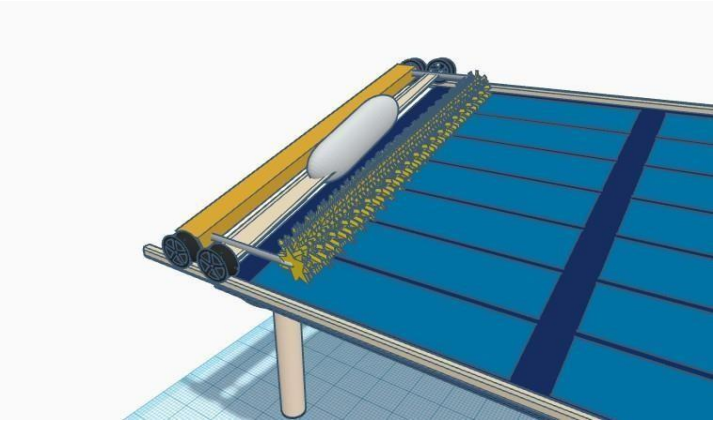


Figure 3.4: Experimental Setup

3.5 Rotating Brush

There are different types of brushes available in the market, each designed for a specific task. For example grout brushes are used for cleaning grout lines between tiles, scrub brushes used for scrubbing tough messes. The type of brush that we have used for our project is called dust brush which has soft Brussels that will avoid any damage to the solar panel while moving along the length of its surface. These Brushes are attached to two 60 RPM geared motors which will control its rotating mechanism.



Figure 3.5: Rotating Brush

3.6 Vacuum Cleaner

The fine particles like dust present on the surface of the solar panel is removed with the help of vacuum suction which collects all the dust and can be discarded as per convenience. To perform efficient cleaning, a 0.2-0.4 hp vacuum is used.



Figure 3.6: Vacuum Cleaner

3.6 Microfiber Cloth



Figure 3.7: Microfiber cloth

The last part in the cleaning mechanism consists of a microfiber cloth that removes any remaining dust particles that is left over after operation of vacuum suction and rotating brush. The use of microfiber cloth will ensure the quality of the surface.

3.7 Ethical considerations

Ethical considerations in designing a solar panel cleaning system involve using eco-friendly

cleaning materials to minimize environmental impact. Ensuring worker safety by implementing proper safety measures and training is crucial. Designing the system to use energy efficiently helps reduce its carbon footprint. Being transparent about how the system works and maintaining it responsibly builds trust. Overall, focusing on these ethical factors ensures the system benefits the environment and people while operating effectively and safely.

Solar panel cleaning systems are crucial for maintaining the efficiency and longevity of photovoltaic (PV) installations. These systems utilize components such as rotating brushes, vacuum systems, and microfiber cloths to remove accumulated dirt and debris that can obstruct sunlight and reduce energy output. Automation is common, with sensors or timers triggering cleaning cycles based on environmental conditions or scheduled intervals. Clean panels can enhance energy production by up to 20%, optimizing ROI over time. Eco-friendly cleaning agents mitigate environmental impact, while safety protocols ensure worker well-being during maintenance operations. Ongoing technological advancements aim to improve cleaning efficiency, reduce water consumption, and integrate smart technologies for enhanced performance and sustainability. Adherence to regulations and standards ensures responsible deployment and operation of these systems in the growing solar energy sector.

3.8 Limitations:

Limitations of the solar panel cleaning system include potential mechanical wear on the panels from brush contact, requiring careful design to minimize abrasion. Weather conditions such as rain or dust storms may affect cleaning effectiveness, necessitating periodic maintenance. Energy consumption by the motors and equipment adds operational costs, and access to remote or elevated installations may pose logistical challenges for regular cleaning and maintenance activities.

CHAPTER 4

RESULTS & DISCUSSION

The team has progressed in designing the solar panel cleaning system. The initial focus was on the rectangular base design. Following a consultation with a mechanical engineering lecture, the team explored the implementation of a vacuum system, including its specifications and design considerations. To gain practical insights, the team visited the college rooftop and interviewed maintenance personnel about cleaning challenges, labor requirements, and solar panel structure. Additionally, discussions were held with a guide to determine the appropriate power supply for the vacuum and battery specifications. Based on the gathered information, the base design has been finalized. Currently, the team is focused on designing and implementing the circuit that will govern the system's movement. The next stage will involve integrating the vacuum system into the overall design.

CHAPTER 5

CONCLUSIONS AND FUTURE DIRECTIONS

5.1 Conclusion

The performance of solar panel is reduced due to the dust and other particles accumulated on it. The fully assembled system was able to remove dust and other particles. Furthermore, it initiated the combination of vacuum cleaner, rotating brush and microfibre cloth to move down and up the panel to clear the dust and other particles. More importantly, the project decreased the daily energy lost compared to the case where the PV panel was left unaltered for many days. Hence implemented automatic cleaning system maximizes the Efficiency.

5.2 Future Directions

The available system can be upgraded into an IoT based system by interfacing different sensors and the gsm of GSM mode can benefit its operation by providing the user the details about its operation and give him control access to intiate or schedule tasks.

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APPENDIX

LIST OF COMPONENTS USED

SL.NO.	COMPONENT	Specification	DETAILS
1	Arduino Uno	-	Microcontroller
2	Booster	MT3608 module	Boosts input voltage
3	Motor Driver	L298N	Drives the DC Motors
4	Ultrasonic sensor	-	Detects the edge of the panel
5	DC Motor	60rpm,30rpm	Rotates brush and the system

LIST OF SOFTWARE TOOLS USED FOR THE PROJECT

SL.NO.	SOFTWARE	VERSION/ DETAILS
1	Proteus	v.8.13
2	Arduino IDE	V2.3.2